



6 (a) State Coulomb's law.

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..... [2]

(b) Fig. 6.1 shows an isolated hollow conducting sphere that is positively charged.

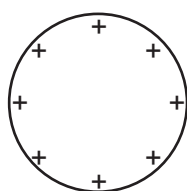


Fig. 6.1

On Fig. 6.1, draw field lines to represent the electric field outside the sphere. [3]

(c) Fig. 6.2 shows the variation of the electric field strength E with distance x from the centre of the sphere in (b).

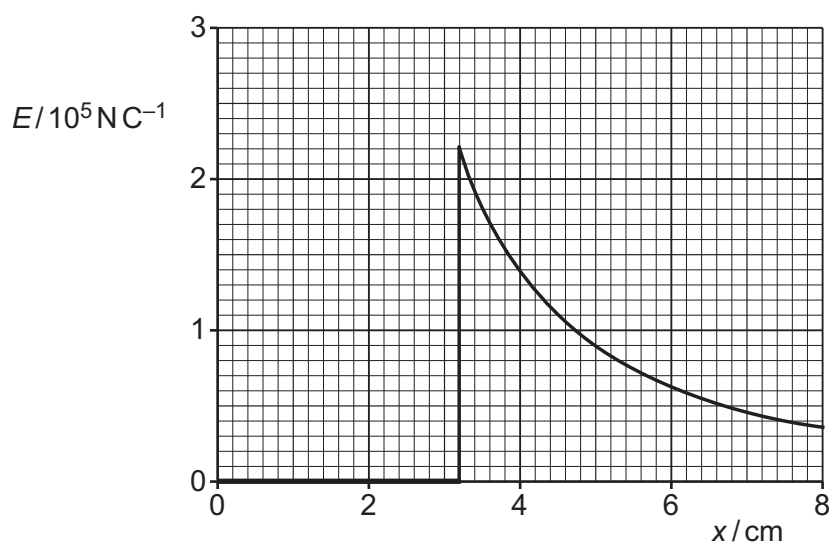


Fig. 6.2





(i) Determine the radius, in cm, of the sphere.

radius = cm [1]

(ii) Calculate the charge on the sphere.

charge = C [3]

(iii) Suggest an explanation for the fact that the electric field inside the sphere is zero.

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..... [1]

[Total: 10]

